Friedrich-Alexander-Universität Erlangen-Nürnberg
Professorship for Open-Source Software

Does Knowing the Speaker Help?

Interview-Grounded Participant Profiles for LLM-Assisted Qualitative Coding in Software Engineering Research

Master's Thesis Proposal

Author: Ahmadreza Nourozi
Programme: M.Sc. Artificial Intelligence
Intended supervisor: Prof. Dr. Dirk Riehle
Data source: Interview studies coded in QDAcity
Planned duration: Six months
Date: October 2026

Abstract

Large language models (LLMs) are increasingly used to code qualitative interview data, but they usually code each segment in isolation. Human analysts instead read the whole interview first and interpret each statement in light of what the participant has said elsewhere. This thesis tests whether giving an LLM a *participant profile*, generated from the other parts of the same interview and grounded in quotations, improves its agreement with human coding of software-engineering interviews. Profiles are built with a two-fold cross-fitting scheme so that the text being coded never contributes to its own profile. To establish that any improvement comes from knowledge of the speaker rather than from additional text, topical cues or general knowledge of the study, real profiles are compared with four matched control profiles, the most important being the profile of another participant of the same study. Replacement-based interventions test whether the model causally relies on the evidence it cites, and a pre-specified mechanism analysis examines when and why profiles help or harm. The thesis reuses interviews for which hundreds of students have each submitted a separate coding in QDAcity with the same codebook. This yields a stable human reference and a full distribution of individual human coders within which the LLM can be placed, while keeping the annotation effort feasible within six months.

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1 Introduction

1.1 Background

Qualitative data analysis is a core method of empirical software engineering. Interviews with practitioners are coded to understand practices, challenges and attitudes that quantitative data cannot capture (Seaman, 1999; Hove and Anda, 2005). Coding is also the most labour-intensive part of such studies: in the study of Shah et al. (2025), two analysts spent a combined 74 hours on a modest set of documents. The Professorship for Open-Source Software develops QDAcity, a web-based tool for this work, and has investigated automated support for coding interview data (Kaufmann et al., 2020; Kaufmann and Riehle, 2019).

Large language models (LLMs) have made such support far more capable. Under deductive conditions, that is, when a codebook is given, several studies report agreement with human coders that approaches human inter-coder reliability (Chew et al., 2023; Törnberg, 2025; Dunivin, 2025; Than et al., 2025). In software engineering, Shah et al. (2025) report Cohen's kappa of about 0.73 for few-shot coding of requirements. What remains unclear is which information the model needs in order to interpret a statement the way an analyst does.

1.2 Problem Statement

LLM coding pipelines typically present a segment, the codebook and, at most, a short description of the study. The rest of the interview is not shown, mainly because of context-window limits and to keep calls independent and reproducible. Human analysts work differently. They first read the whole transcript, and they interpret a statement such as "that was the real problem for us" by knowing what "that" refers to, what role the participant holds and what the participant has said about the topic before. In the follow-up interviews of Moats and Pretnar Žagar (2026), annotators reported that much of the difficulty of coding isolated statements came from not knowing who was speaking, and the authors conclude that decontextualised coding is artificial for humans and machines alike.

This raises a simple but untested question: does an LLM code an interview segment more like the human reference when it knows what the same participant has said in the rest of the interview? Answering it rigorously requires three things that existing work does not provide. First, the speaker information must be constructed so that the segment being coded cannot leak into it. Second, any improvement must be separated from mere effects of adding text, topical keywords or general knowledge of the study. Third, it must be shown that the model actually uses the speaker information it is given, and in which situations this helps or harms.

1.3 Purpose and Scope

The purpose of this thesis is to test whether interview-grounded participant profiles improve LLM-assisted deductive coding of software-engineering interviews, whether any improvement is specific to the speaker, whether the model causally relies on the profile evidence, and through which mechanisms profiles help or harm. The available data make this possible with unusual rigour: in the professorship's teaching, the same interview transcripts are coded separately by hundreds of students with a shared codebook, so that human agreement can be described as a distribution rather than estimated from two coders. The thesis deliberately concentrates on these questions. Prompt engineering, reasoning modes, inductive coding, fairness analysis and the development of new QDAcity features are outside its scope (Section 6). The work is nevertheless closely tied to QDAcity, since it uses interview data already coded in QDAcity and its results inform how future AI-assisted coding in QDAcity should present context.

2 Theoretical Framework: Why Speaker Context Should Matter

The central assumption of the thesis, that knowing the speaker improves coding, is intuitive but needs a theoretical basis. This section derives it from established accounts of interpretation and qualitative practice and turns it into testable mechanisms.

2.1 Indexicality and the Two Contexts of Interpretation

Ethnomethodology describes everyday talk as *indexical*: the meaning of an utterance depends on the circumstances of its production, including who produces it, and cannot be fixed by its words alone (Garfinkel, 1967). Cicourel (1964) drew the methodological consequence that assigning standard categories to fragments of talk requires the analyst to suppress the context that makes these fragments different. Moats and Pretnar Žagar (2026) distinguish two contexts that are lost when coding isolated snippets: the *situational* context of the surrounding conversation, and the *biographical* context of who the speaker is, including their expertise and earlier positions. A participant profile is an attempt to restore the second without simply restoring the first.

2.2 The Part and the Whole in Qualitative Analysis

Interpretive analysis proceeds in a circle: parts are understood through the whole, and the whole through its parts. Qualitative methods make this explicit. Thematic analysis begins with familiarisation, in which the analyst reads the entire data set before generating any code (Braun and Clarke, 2006), and coding manuals advise reading a case as a whole before coding its parts (Saldaña, 2021). Research interviews are, moreover, co-constructed narratives in which later answers build on earlier ones (Brinkmann and Kvale, 2015). Segment-wise LLM coding skips the familiarisation step entirely. A participant profile can be understood as a compact, explicit substitute for it, and the success of holistic, author-level reading in LLM text analysis (Benoit et al., 2026) supports this view.

2.3 Why Software-Engineering Interviews Are Especially Context-Dependent

Software-engineering interviews ask practitioners about their own practices, projects and organisations (Hove and Anda, 2005). Their answers are full of references to entities introduced earlier (“the pipeline”, “that migration”, “our old process”), and evaluations depend on the participant’s role and experience: “the review process is fine” means something different from a team lead who designed the process than from a newcomer who struggles with it. Codes commonly used in such studies, for example challenges, motivations, practices and attitudes towards tools, therefore often require knowing the participant’s perspective.

2.4 Why a Compact Profile Rather Than the Full Transcript

An obvious alternative is to give the model the whole transcript. Long inputs, however, are used unevenly by LLMs, which tend to neglect information in the middle of long contexts (Liu et al., 2024), and a full transcript also restores the local situational context, which makes it impossible to tell which kind of information helped. A compact, grounded profile isolates biographical information, keeps the input short and inspectable, and makes it possible to trace a decision back to specific evidence. The full transcript is therefore used in this thesis as a reference condition rather than as the method itself.

2.5 Expected Mechanisms

From these accounts, four mechanisms are derived through which a profile can help, and one through which it can harm. They are pre-specified and tested in RQ4.

M1 Reference resolution. The segment refers to something explained elsewhere in the interview, and the profile makes the referent clear.

M2 Evaluative disambiguation. Hedging, irony or mixed valence make the segment's stance unclear; the participant's established attitude resolves it.

M3 Perspective framing. The meaning of an evaluation depends on the participant's role or experience, which the profile supplies.

M4 Narrative continuity. The segment continues an account or argument started elsewhere.

H1 Over-projection (harm). The model projects the participant's general views onto a segment that expresses something different, or is distracted by evidence that is topically related but irrelevant.

3 Literature Review

3.1 LLM-Assisted Qualitative Coding

Evidence that LLMs can apply a codebook has accumulated quickly (Chew et al., 2023; Törnberg, 2025; Dunivin, 2025), although results vary substantially across code types (Than et al., 2025). Shah et al. (2025) provide the closest software-engineering reference: in requirements documents, few-shot prompts with rich context about the described system reached substantial agreement with human analysts. Their context manipulation concerned the system, not the people, and requirements are documents in which the speaker plays almost no role. Interviews are the opposite case.

3.2 The Missing Speaker

Moats and Pretnar Žagar (2026) provide the most direct evidence for the gap addressed here. Coding values in parliamentary speeches, their annotators preferred human over LLM labels but agreed poorly with one another (Cohen's kappa between 0.12 and 0.25). Interviews with the annotators showed that they drew on linguistic nuance, cultural knowledge and knowledge of the speaker, whereas the models assigned the maximum number of codes and attended to keywords even in subordinate clauses. The authors argue that experimental designs force humans to "meet the machines half-way" by coding isolated sentences. Related work points in the same direction: Benoit et al. (2026) obtain their best results when the model first summarises an author's overall position, and Ashwin et al. (2025) show that LLM coding errors correlate with characteristics of the speaker. To the author's knowledge, no study has supplied an LLM coder with grounded information about the speaker drawn from the same interview and measured its effect.

3.3 Grounded Profiles and Personas

Rahimzadeh et al. (2025a) generate natural-language user profiles from large volumes of posts and find that extractive profiles, which quote evidence, outperform abstractive ones. SYNTHIA checks every claim of a generated persona against the source material as supported, contradicted or unverifiable (Rahimzadeh et al., 2025b). De Paoli (2026) derives personas from qualitative interviews with LLMs, but as archetypes for design, not as context for coding individual statements. Borchers et al. (2025) give personas to the *coding agents* and observe minimal accuracy gains; in their design, the speaker of the coded text remains unknown.

3.4 Evaluating LLM Coding

Agreement with human coders is not the same as quality: a blind expert may prefer LLM codes where human consensus encodes a shared bias (Liu et al., 2026). When several human coders are available, the alternative annotator test provides a statistical procedure for judging whether an LLM can stand in for one of them (Calderon et al., 2025). Human labels also vary legitimately, and recent work argues that this variation should be modelled as information rather than collapsed into a single label (Plank, 2022; Uma et al., 2021). When many annotators of varying reliability label the same items, their labels

can be aggregated by majority vote or by models that estimate each annotator's error rates (Dawid and Skene, 1979). Choices of model and prompt can change conclusions, which calls for pre-registration and robustness checks (Baumann et al., 2025; Ruler and Benoit, 2026). For explanations, faithfulness research shows that stated reasons need not reflect the factors that drive a decision (Turpin et al., 2023), and that faithfulness should be tested through interventions on the cited evidence, measuring both whether removing it changes the decision and whether it alone suffices (DeYoung et al., 2020; Atanasova et al., 2023).

3.5 Research Gap

Existing work shows that LLMs can code under deductive conditions, that missing speaker context limits coding of isolated statements, and that grounded profiles can be generated and checked. It does not show whether speaker information from the same interview improves LLM coding, whether such an effect is specific to the speaker rather than to added text or topic, whether the model actually relies on the speaker evidence, or why. This thesis addresses these four questions.

4 Research Questions and Hypotheses

4.1 Research Questions

Main question. Does a profile generated from a participant's other interview statements improve LLM-assisted deductive coding of software-engineering interviews, compared with coding without speaker context?

RQ1 *Effect.* Do extractive and abstractive participant profiles increase agreement with the human reference compared with no profile, and what share of the improvement obtainable from the full transcript do they achieve?

RQ2 *Speaker specificity.* Is the improvement caused by information about the specific participant, rather than by added text, topical keywords or general knowledge of the study?

RQ3 *Causal reliance.* When a profile changes a coding decision, does the decision causally depend on the profile evidence the model cites?

RQ4 *Mechanisms.* For which segments, and through which of the mechanisms M1 to M4 and H1, do profiles help or harm?

4.2 Hypotheses

The hypotheses follow from Section 2 and will be pre-registered before the main experiment.

H1 Both profile forms increase agreement over no profile and move the model upwards in the distribution of individual human coders; the extractive form performs at least as well as the abstractive form (Rahimzadeh et al., 2025a).

H2 The real profile outperforms each of the four control profiles. In particular, it outperforms the profile of another participant of the same study, and this advantage shrinks as the other participant's views become more similar to those of the real participant.

H3 Replacing the evidence the model cites changes its decisions significantly more often than replacing the same amount of uncited evidence.

H4 Profile gains are concentrated in units that show indicators of mechanisms M1 to M4 and on which human coders disagree more; harms occur mainly through over-projection (H1).

4.3 Definitions of Key Terms

Unit. The unit of coding: a paragraph-level part of one participant answer, presented together with the interviewer question it answers.

Question block. One main interviewer question together with the participant's answer and any follow-up exchanges.

Participant profile. A text describing a participant, generated only from that participant's question blocks in the other fold of the same interview. The *extractive* form is a fixed number of quotations; the *abstractive* form is a short summary of recurring perspectives, experiences, concerns and reasoning patterns in which every statement cites supporting quotations.

Human reference. For every unit and code, the majority decision of the separately submitted student codings, together with the proportion of coders who assigned the code (soft label). The reference is treated as an imperfect standard, not as ground truth.

Human-human agreement baseline. The distribution of agreement between each individual coder and the majority of all other coders, used to interpret LLM agreement.

Reliable and contested decisions. A unit-code decision is reliable if the share of coders who assigned the code differs significantly from one half, and contested otherwise.

Control profile. A text placed in the profile slot that matches the real profile in length, format, topic or source study but carries no information about the specific participant.

5 Methodology

5.1 Research Design Overview

The thesis is a controlled within-unit experiment: every unit is coded under every condition, so that conditions are compared on identical material. Figure 1 summarises the design. Existing human codes form the reference; profiles are built from the same interviews by cross-fitting; an LLM codes each unit under the experimental and control conditions; and four analyses answer the four research questions. A development set of 20% of the participants is used to fix the prompt, the primary profile form and the analysis plan, which is then pre-registered; all reported results come from the remaining 80%.

5.2 Data: Existing Human-Coded Interviews

Source. The thesis reuses interview transcripts that have already been coded by humans in QDAcity in the professorship's teaching, notably the course *Nailing Your Thesis*. In this course, every student receives the same interview transcripts and the study's codebook, reads each transcript and assigns codes from the codebook to its passages. Each transcript thereby receives hundreds of separately submitted student codings, and the procedure has been repeated over several semesters. Reusing these codings is an instance of secondary analysis of qualitative data (Heaton, 2008). It removes the main cost of comparable studies, which must produce human codes from scratch (Shah et al., 2025), and it provides something such studies cannot: a large number of separately submitted human codings of exactly the same text.

Selection criteria. A set of interviews coded in one semester with one codebook forms a *study*. A study is included if (i) the consent and terms under which the interviews and the student codings were collected allow secondary research use, or the necessary approval can be obtained; (ii) each interview contains enough material for two folds (at least six question blocks besides the introductory block); and (iii) the language is German or English. Because each study contributes only a small number of interviewees, studies from several semesters are combined to increase the number of distinct participants, with a target of 25 to 35 participants and roughly 1,500 to 3,000 units. Each study is coded with its own codebook, and codebook is modelled as a random effect.

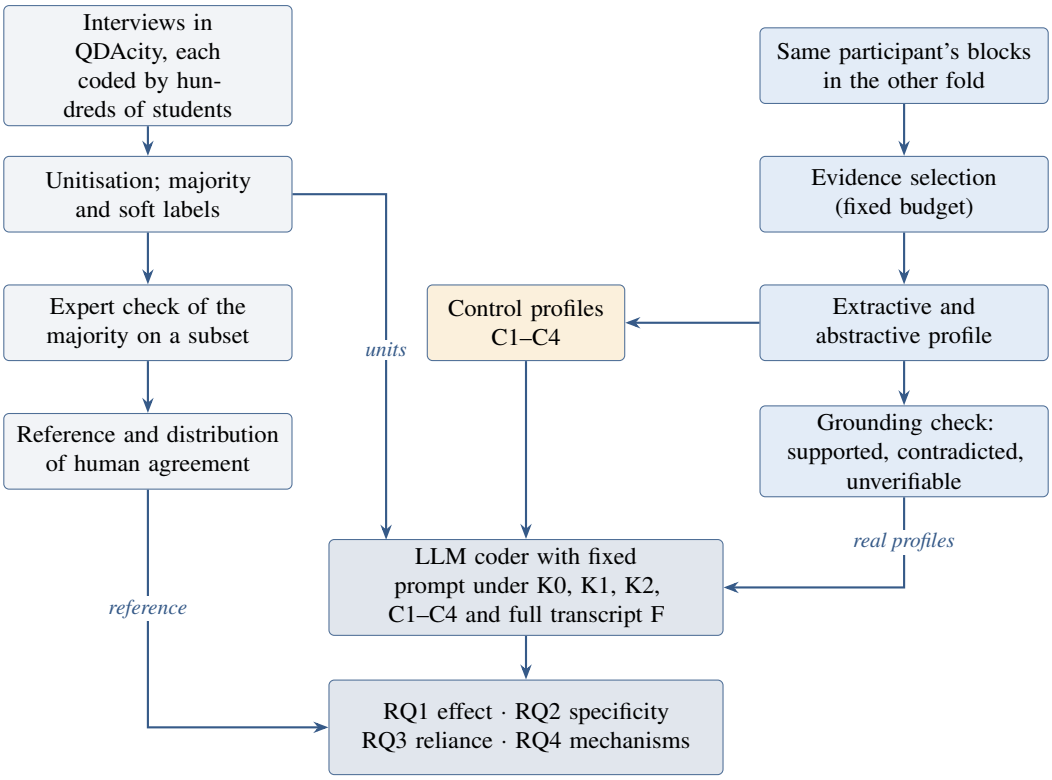


Figure 1: Research design. The reference path (left) and the profile path (right) are built independently and meet only in the evaluation.

Coder inclusion. All complete codings are used. A coding is excluded only if it covers less than a pre-registered share of the transcript. If a reference solution was discussed in class before the submission deadline, only codings submitted before that discussion are used, so that the remaining codings come as close as possible to independent judgements.

5.3 Units and Reference Labels

Unitisation. Coded spans in QDAcity vary in length and boundary, which is a known source of disagreement in interview coding (Campbell et al., 2013). The thesis therefore defines fixed units: paragraph-level parts of participant answers of at most about 150 words, never crossing a speaker turn. A code is assigned to a unit if a human code span covers at least half of the unit or at least half of the span lies within the unit. This rule is applied identically to all coders, so that unitisation does not differ between humans.

Majority and soft labels. For every unit and code, the proportion p of coders who assigned the code is computed. The hard reference is the majority decision ($p \geq 0.5$), and p itself is kept as a soft label. With hundreds of coders, a binomial test separates *reliable* decisions, whose majority differs significantly from one half, from *contested* decisions, on which coders are split. Hard-label metrics are computed on reliable decisions; contested decisions are evaluated with soft-label metrics, in line with work that treats human label variation as information rather than noise (Plank, 2022; Uma et al., 2021). As a robustness check, the majority is replaced by a Dawid–Skene aggregation that weights coders by their estimated reliability (Dawid and Skene, 1979).

Distribution of human agreement. For each individual coder, agreement with the majority of all *other* coders is computed. The resulting distribution over hundreds of coders is the human-human agreement baseline. The LLM's agreement with the majority is located within this distribution, so that each condition

can be reported as a percentile, for example whether a profile moves the model from below to above the median coder. Because the coders are students, this percentile describes the model relative to trained novices rather than experienced researchers. The alternative annotator test (Calderon et al., 2025) is additionally applied, with random samples of coders per interview, to ask whether the LLM, with or without a profile, can statistically stand in for a human coder.

Expert check. A majority of novices may still share systematic misreadings (Liu et al., 2026). The author and a second researcher therefore independently code a stratified sample of about 150 units, adjudicate their disagreements and compare the result with the student majority. If an instructor's reference coding exists for a study, it is used for this check instead. The check requires roughly eight to ten hours per researcher.

5.4 Interview-Grounded Participant Profiles

Cross-fitting split. Each interview is divided into question blocks (Figure 2). The introductory block, in which participants typically describe their role and experience, is never coded in the experiment and is available to both profiles, because it carries the biographical information that the theory identifies as relevant. The remaining blocks are assigned alternately to fold A and fold B, which balances interview phase and topic between the folds. The profile for coding units in fold B is generated from the introductory block and fold A, and vice versa. A unit therefore never contributes to its own profile.

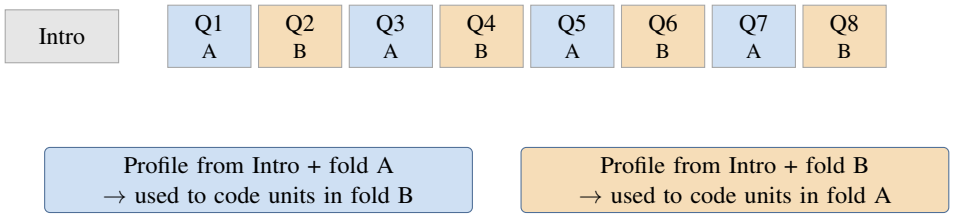


Figure 2: Two-fold cross-fitting within one interview. Blocks alternate between folds; the introductory block is shared by both profiles and is never coded.

Alternation means that a block adjacent to a target unit may contribute to its profile, which could reintroduce local situational context. Because a profile is generated once per participant and fold, independently of any target, this is limited; a pre-registered robustness analysis nevertheless repeats all results without the units whose adjacent block is quoted in the profile.

Fixed evidence budget. All profiles have the same size: the extractive profile consists of exactly ten quotations of at most 60 words each, and the abstractive profile of at most 200 words. Quotations are selected to cover the participant's recurring perspectives across different interview questions, using a multilingual embedding model (Chen et al., 2024) with a diversity criterion. Because every participant contributes a full interview, profile size no longer depends on how much external activity a person has, and differences in profile quality are reduced to differences in content.

Generation and grounding check. Profiles are generated by a locally run LLM that sees neither the codebook nor any human codes, so that profiles cannot contain code labels. The abstractive profile summarises recurring perspectives, experiences, concerns and reasoning patterns, and every statement cites the quotations it rests on. Each statement is then checked against its quotations as supported, contradicted or unverifiable (Rahimzadeh et al., 2025b); unsupported statements are removed, a sample is checked manually, and the share of supported statements is reported.

5.5 Experimental Conditions and Controls

All conditions share the same prompt: codebook definitions, a short description of the study and its interview guide, the unit and the interviewer question it answers. They differ only in the content of a single profile slot (Table 1). The four controls are constructed for the primary profile form, which is chosen on the development set and fixed in the pre-registration.

Table 1: Experimental conditions. All conditions use the same prompt and differ only in the content of the profile slot.

Condition	Content of the profile slot	Comparison with the real profile isolates
K0 No profile	Empty	Baseline
K1 Extractive	Ten quotations from the participant's other fold	—
K2 Abstractive	Grounded summary of the participant's other fold	—
C1 Length	Neutral text of identical length	Effect of added text
C2 Other participant	Profile of another participant of the same study, built identically	Effect of the specific person, with topic, format and study held constant
C3 Composite	Same budget of quotations drawn from several other participants of the same study	Effect of general knowledge of the study's discourse
C4 Keywords	Content words of the real profile in random order, identical length	Keyword priming
F Full transcript	The whole interview, with the target unit marked	Reference: same information as human coders

Control C2 is the decisive test. Because all participants of a study answered the same interview guide, a profile from another participant contains the same topics, vocabulary and format, but describes a different person. If the real profile outperforms C2, the improvement cannot be explained by topical cues alone. Control C3 additionally separates knowledge of the study's typical discourse from knowledge of the individual.

5.6 Coding Procedure

The coding model is an open-weight model from the Qwen3 family (Qwen Team, 2025), run locally in non-thinking mode at temperature zero. Prompt wording is fixed on the development set and not varied in the experiment, since the thesis does not study prompt design. For each unit the model returns, in a constrained JSON format, the assigned codes, the identifiers of any profile quotations it relied on, and a confidence score. To guard against conclusions that depend on a single model or run (Baumann et al., 2025; Ruler and Benoit, 2026), the main contrasts are replicated with a model from a second open-weight family and with three sampled runs. All processing takes place on university hardware, so that interview data never leave the university. The experiment requires roughly 40,000 model calls, which is feasible within an estimated 100 GPU hours.

5.7 RQ1: Effect of Profiles

Agreement between the model and the human reference is measured per code with F1, across codes with macro-F1 and Jaccard similarity, and with Krippendorff's alpha (Krippendorff, 2018). Each condition is also reported as a percentile within the distribution of individual human coders (Section 5.3), and soft-label agreement is measured as the Brier score between the model's confidence and the proportion of coders who assigned the code. The primary contrasts are K1 versus K0 and K2 versus K0. In addition, the *gap closure ratio*,

$$\mathrm{GCR} = \frac{A_{\text{profile}} - A_{\text{K0}}}{A_{\text{F}} - A_{\text{K0}}},$$

where A denotes agreement, expresses which share of the improvement obtainable with the full transcript the compact profile achieves. A ratio close to or above one would mean that the profile preserves the useful information of the transcript at a fraction of its length.

5.8 RQ2: Speaker Specificity

Speaker specificity is claimed only if the real profile outperforms *each* of the four controls. This is tested as an intersection-union test: each of the four one-sided contrasts (real versus C1, C2, C3 and C4) is tested at the 5% level, and the overall claim holds only if all four are significant. An intersection-union test needs no multiplicity correction and avoids the selection bias that comparing against the post-hoc "best" control would introduce (Berger, 1982). In addition, a gradient analysis exploits control C2: the similarity between two participants is measured by the overlap of the codes humans assigned to them, and H2 predicts that the disadvantage of a swapped profile grows as this similarity decreases. Such a dose-response pattern would be difficult to explain by anything other than speaker-specific information.

5.9 RQ3: Causal Reliance on Profile Evidence

Deleting evidence and observing whether a decision changes is a weak test, because evidence may be redundant and deletion alters prompt length and structure. The thesis therefore uses replacement-based interventions that keep length and structure constant (Table 2). They are applied to all *profile-sensitive decisions*, that is, unit-code decisions that differ between K0 and the primary profile condition.

Table 2: Interventions for testing causal reliance on cited profile evidence.

Intervention	Construction	Interpretation
I1 Cited replacement	The quotations the model cited, together with near-duplicates (embedding similarity above a fixed threshold), are replaced by length-matched quotations of another participant answering the same interview question	Necessity of the cited evidence, robust to redundancy
I2 Random replacement	The same number of uncited quotations is replaced in the same way	Baseline for any perturbation of the profile
I3 Sufficiency	Only the cited quotations are kept; all others are replaced as in I1	Whether the cited evidence alone suffices
I4 Counterfactual edit	For a sample of about 100 decisions, a cited quotation is minimally edited so that it implies the opposite, and the edit is checked manually	Whether the decision follows the content of the evidence

Following the comprehensiveness and sufficiency logic of DeYoung et al. (2020) and the intervention-based faithfulness tests of Atanasova et al. (2023), causal reliance is supported if decisions flip significantly more often under I1 than under I2, are mostly preserved under I3, and change in the predicted direction under I4. Since even these interventions cannot prove internal mechanisms, the results are reported as evidence consistent with, or against, causal reliance.

5.10 RQ4: Mechanism Analysis

RQ4 asks why profiles help or harm, in two complementary steps.

Mechanism indicators. Before any experimental output is inspected, every unit is annotated for indicators of the mechanisms in Section 2: unresolved references to earlier talk (M1), hedging, irony or mixed valence (M2), evaluations whose meaning depends on role or experience (M3), and continuation of an earlier account (M4). Annotation is pre-labelled automatically and validated by two human raters on

a stratified sample of 200 units; indicators with Cohen's kappa below 0.6 are merged or dropped. Two objective measures complement them: unit length and the disagreement among the hundreds of human coders, measured as the entropy of their decisions. Human disagreement is the most direct available measure of how open a unit's interpretation is. These indicators replace a subjective judgement of whether a unit is "speaker-dependent" with observable properties that can be checked for reliability. Mixed-effects models then test whether the profile gain is larger for units with these indicators.

Analysis of changed decisions. For a random sample of about 100 decisions that a profile corrects and about 100 that it worsens, two raters read the unit, the profile and the cited evidence and classify the change by mechanism (M1 to M4 for corrections, H1 or other for errors), with inter-rater agreement reported. Combined with the causal tests of RQ3, this shows not only whether profiles help, but through which kind of information they do so.

5.11 Statistical Analysis and Pre-Registration

The basic observation is a unit-code decision under one condition. Because every unit is coded under every condition, observations of the same unit are repeated measures and are not independent. Correctness relative to the reference is therefore modelled with mixed-effects logistic regression, with condition as fixed effect and random intercepts for the unit-code decision and for the unit, nested within participant and study, together with a crossed random intercept for code:

$$\operatorname{logit} P(\text{correct}_{ijc}) = \beta_0 + \beta^\top \mathbf{x}_c + u_{ij} + v_i + w_{p(i)} + s_{d(i)} + g_j,$$

where i indexes units, j codes, c conditions, $p(i)$ and $d(i)$ the participant and study of unit i, u_{ij} the unit-code intercept and v_i, $w_{p(i)}$, $s_{d(i)}$, g_j the intercepts for unit, participant, study and code. If the model does not converge, the random-effects structure is simplified in a pre-registered order, keeping the unit-level terms that represent the repeated measures. Confidence intervals are obtained with a cluster bootstrap over participants, which preserves all observations of a unit together. The primary contrasts of RQ1 are corrected with the Holm procedure; RQ2 uses the intersection-union test described above. Because each interview is coded by many students but the number of distinct participants is limited, participant-level uncertainty is the binding constraint. The number of interviews needed is determined by a simulation-based power analysis using effect sizes and variance components estimated on the development set. If fewer than about 20 distinct participants are available, participant is treated as a fixed effect while the unit-level random intercepts are retained, uncertainty is estimated by resampling question blocks within interviews, and generalisation claims are limited accordingly. Hypotheses, primary contrasts, the primary profile form and all exclusion rules are pre-registered before the test set is coded.

5.12 Validity

Construct validity. The reference is a majority of novice coders; the expert check, the Dawid–Skene robustness analysis and the separation of reliable and contested decisions make its limitations visible rather than hiding them. *Internal validity.* Human coders saw the whole interview, whereas the model sees the unit and a profile. This information mismatch affects the absolute level of agreement, but not the comparisons between conditions, which all use the same reference and differ only in the profile slot. The full-transcript condition F matches the information available to human coders and therefore serves as the reference point for how much of the gap a profile closes. Accordingly, the thesis claims improved agreement with the human reference, not human-equivalent coding. Leakage is prevented by cross-fitting and tested by the adjacency robustness analysis. *External validity.* Results are limited to the selected studies and codebooks; including several studies and modelling codebook variation mitigates this. Interviews used in teaching may have been chosen for their clarity, so results may not fully transfer to typical research interviews. *Conclusion validity.* Mixed-effects models, the cluster bootstrap, pre-registration and replication across models and runs protect against spurious findings.

6 Scope, Assumptions and Limitations

Delimitations. To keep the thesis focused and feasible within six months, the following are explicitly out of scope: systematic prompt engineering, since one prompt is fixed on the development set; comparison of reasoning and non-reasoning modes; inductive coding, whose open-ended nature conflicts with evaluation against a fixed codebook; fairness analysis across participant groups; development of new QDAcity features; and profiles built from external sources such as GitHub activity. No model is fine-tuned, and no commercial API is used.

Assumptions. The study assumes that the majority of hundreds of separately submitted student codings, checked against expert coding, is an adequate though imperfect reference; that the introductory and remaining blocks of an interview carry information about the participant that is relevant to coding other blocks; and that codebooks of the selected studies are applied consistently enough to be used deductively.

Limitations. Profiles capture only what participants say in one interview. A within-interview profile can never be entirely free of situational context, which is why the controls, the full-transcript reference and the adjacency analysis are needed. Results obtained with open-weight models of moderate size may not transfer to the largest commercial models. Finally, the student codings were submitted separately, but the students attended the same course, received the same instructions and may have discussed the transcripts with one another. Their independence is therefore an approximation rather than a literal statistical assumption. Accordingly, the binomial threshold that separates reliable from contested decisions is used as a descriptive cut-off rather than as a formal significance statement, and the results are checked for robustness under a stricter threshold.

7 Ethical and Data-Protection Considerations

Interview transcripts are personal data, and the codes from courses are student work. Before any analysis, the consent and terms under which the selected data were collected are checked with the supervisor, and only data whose secondary research use is permitted, or for which the necessary approval is obtained, are included. All processing takes place on university infrastructure with locally run models, in accordance with the GDPR and FAU's data-protection requirements; no data are sent to external services. Participants and coders are pseudonymised, profiles are used only as model input and are never published, and results are reported in aggregate. Student codings are coursework: no result is reported about an individual student, and the percentile analysis uses only the distribution of coders. Transcripts and profiles are not released; the replication package contains code, prompts, configurations and aggregate results.

8 Expected Contributions

1. The first controlled test of whether speaker context from the same interview improves LLM-assisted coding of software-engineering interviews.
2. The first evaluation that places an LLM coder, with and without speaker context, within the distribution of hundreds of separately submitted human codings of the same interviews.
3. A method for interview-grounded participant profiles with cross-fitting, a fixed evidence budget and a grounding check.
4. A control design, combining a same-study swapped profile with an intersection-union test, that separates speaker-specific information from added text and topical cues and can be reused in any study that adds context to LLM prompts.
5. A replacement-based protocol for testing whether an LLM causally relies on the evidence it cites.

6. Evidence on the mechanisms through which speaker context helps or harms, and implications for how AI-assisted coding in QDAcity should present context to analysts.
7. A pre-registration and replication package.

9 Work Plan

The thesis is planned for six months (Figure 3). Four gates mark points at which progress is reviewed with the supervisor.

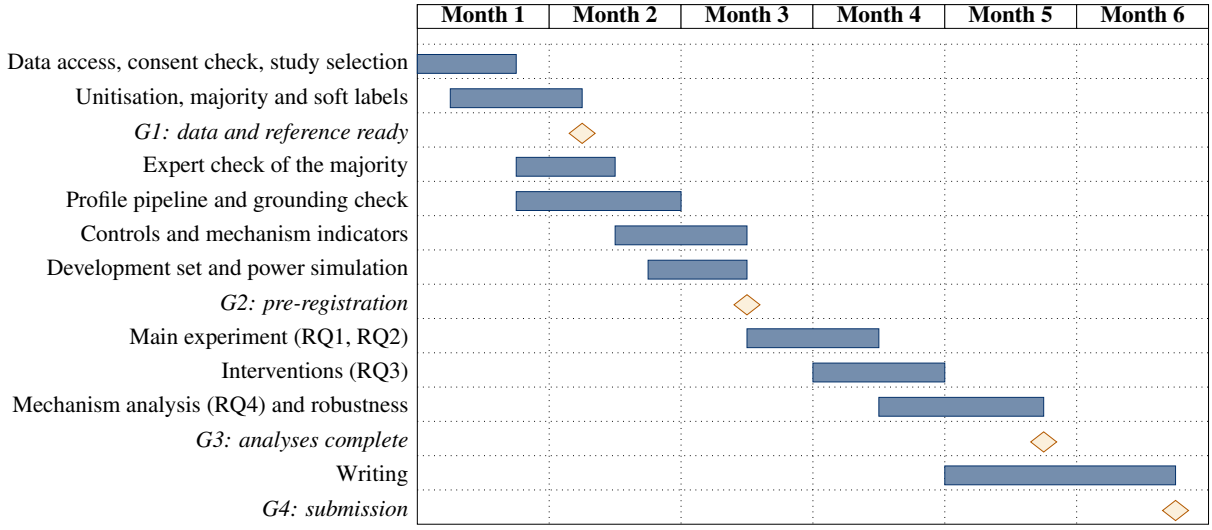


Figure 3: Six-month work plan with four review gates (one column corresponds to about one week).

10 Risks and Mitigation

Table 3 lists the main risks identified at the proposal stage.

Table 3: Main risks and planned mitigation.

Risk	Mitigation
Secondary use of data not covered by consent	Clarified in month 1; only permitted studies are used; interview studies of the professorship with explicit research consent serve as fallback.
Few distinct participants per study	Studies from several semesters are combined; below about 20 participants, the pre-registered fallback analysis treats participant as a fixed effect and limits generalisation claims.
Codebooks differ between semesters	Each study is coded with its own codebook; codebook is modelled as a random effect.
Systematic novice misreadings	Expert check of about 150 units; Dawid–Skene aggregation; separate evaluation of reliable and contested decisions.
Student codings not fully independent	Only codings submitted before any in-class discussion of a reference solution are used.
Interviews too short for two folds	Inclusion criterion of at least six question blocks besides the introduction.
Full transcripts exceed the context window	Long-context models; condition F reported only for interviews that fit, with the number stated.
Small or null effect	Power simulation and pre-registration make a null result interpretable and publishable.

11 Preliminary Thesis Outline

1. Introduction
2. Theoretical framework: speaker context in qualitative interpretation
3. Related work: LLM-assisted coding, profiles and evaluation
4. Research design and data
5. Interview-grounded participant profiles and control conditions
6. Results: effect and speaker specificity (RQ1, RQ2)
7. Results: causal reliance and mechanisms (RQ3, RQ4)
8. Discussion, implications for QDAcity and threats to validity
9. Conclusion and future work

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